Simplifying assumptions in power analysis

A replication of Kupzyk's longitudinal cluster trial study

Abstract

We partly replicated the planning findings of Kupzyk (2011), who reported that treatment imbalance and attrition cost more power than unequal cluster sizes, that losses were greatest at the lowest intraclass correlation, that up to eight extra clusters were needed, and that empirical power was usually lower than power calculated at the generating covariance. Exact replay requires the original software state, simulation streams, and dropout masks. The mean loss for 70/30 allocation was 6.409 versus 6.443 percentage points (published versus ours), and 15% interval dropout cost 4.391 versus 4.472. The maximum extra count was eight versus seven, and empirical power was lower in 41 versus 40 of 54 cells. All 54 initial empirical powers were inside the adjusted comparison bands. Three sample-size crossings moved by one cluster, and the extreme-setting Optimal Design rejection rates differed.

Table 1. Headline comparisons. Published findings and contrasts come from Kupzyk's Tables 1, 4 and 5 and pp. 106–107. Our values come from the R reconstruction.

Contrast	Published	Ours	Signed effect		Effect magnitude	Direction	Verdict
			Source	Ours			
5% dropout	1.3 pp mean loss	1.27 pp mean loss	1.327	1.275	smaller	same direction	match
15% dropout	4.4 pp mean loss	4.47 pp mean loss	4.391	4.472	larger	same direction	match
70/30 allocation	6.4 pp mean loss	6.44 pp mean loss	6.409	6.443	larger	same direction	match
Normal sizes	0.45 pp mean loss	0.44 pp mean loss	0.454	0.438	smaller	same direction	match
Skewed sizes	0.85 pp mean loss	0.78 pp mean loss	0.851	0.778	smaller	same direction	match
ICC .05 vs .15 at 15% dropout	All peak at .05. Skew rises .61 to .67 pp despite reported decline.	All peak at .05. Skew loss rises .56 to .62 pp.	3.037	3.062	larger	same direction	partial
Maximum extra clusters	Up to 8 extra clusters	Up to 7 extra clusters	8	7	smaller	same direction	partial
Count below minus 27	41 of 54 below	40 of 54 below	14	13	smaller	same direction	match

A positive effect denotes power lost versus the matched reference (percentage points, pp), greater dropout loss at ICC .05 than .15 (pp), extra clusters versus the between–within reference, or a count above half the cells, respectively. The rounding tolerances are 0.001 pp for the five main losses, 0.002 pp for the ICC contrast and 0.5 for integer counts. These point estimates do not establish equivalence. Verdicts address the scientific patterns. Monte Carlo, mask and reading uncertainty are assessed separately.

1 Introduction

We partly replicated Kupzyk's headline findings. The estimates cover new samples from the reported design. Exact recovery of the original fits and realized dropout masks is unavailable. In the final planning grid, 70/30 allocation cost 6.409 versus 6.443 percentage points (published versus ours), and 5% and 15% interval attrition cost 1.327 versus 1.275 and 4.391 versus 4.472. Unequal sizes had smaller mean costs, and all five mean losses peaked at an intraclass correlation (ICC) of .05. Skewed sizes had a higher loss at .15 than at .10 in both analyses. We matched 51 of 54 cluster thresholds exactly. The maximum

increase was eight versus seven clusters, and empirical power fell below exemplary power in 41 versus 40 of 54 cells.

The planning question is how much power a trial loses when convenient sample-size assumptions fail. Kupzyk (2011, pp. 1–6, 64–67) compared three methods for a treatment-by-time interaction in longitudinal cluster trials. Formulaic power applies an analytic expression to a simplified design. Exemplary power uses the covariance and layout of a specified design. Empirical power is the fraction of repeated fitted trials that reject the null. These quantities are especially informative together when dropout changes the layout or finite-sample tests use different reference degrees of freedom.

2 Power losses and sample-size consequences

Allocation imbalance had the largest mean power cost in both final exemplary grids, followed by high attrition. In those grids, the mean losses for normal and skewed cluster sizes were .454 and .851 pp in the published data and .438 and .778 pp in the reconstruction. Each contrast holds the other factors fixed. This gives 27 matched pairs for allocation and 18 for each attrition or size contrast. These are averages over the experimental settings. For example, one normal-size pair changes sign between the published and reconstructed final grids, and five do so in the initial empirical grid. In initial empirical cells 16 and 17 at ICC .05, the equal-minus-normal size contrast was −.4183 pp in the source and −.11328 pp in R, a small power gain for normal sizes in that pair. The small mean size effects describe the tested size distributions. The ranking applies to these four-occasion Gaussian designs and the specified dropout mechanism. See Kupzyk (2011, Tables 1–2 and 4) and `results/full/paired-contrasts.csv`.

The published hypothesis placed attrition first, while the published results and the reconstruction place allocation first. The final-grid comparison is 6.409 versus 4.391 pp in the source and 6.443 versus 4.472 pp in R for allocation and 15% attrition, respectively. The mean ordering holds when each final exemplary contrast changes just one factor, averaging over pairs matched on the other factors. The evidence explains the discrepancy as a result of the specific changes compared in this design (`results/full/comparison-summary.json`).

Disagreement record `predicted-order`.

The lowest ICC had the greatest loss for each violation, with one exception to a strictly declining trend across ICC. The skewed-size loss rises from .6098 to .6673 pp between ICC .10 and .15 in the source, and from .5614 to .6248 pp in R. For 15% interval dropout, the loss at .05 exceeds the loss at .15 by 3.037 pp in the source and 3.062 pp in R. The source ordering at the lowest ICC is supported, and the monotonic generalization in the discussion fails under recomputation from Table 1 (`results/source-identities.json`).

Disagreement record `skew-icc-monotonicity`.

2.1 How many clusters restore 80% power?

The first threshold for mean exemplary power matched the published threshold in 51 of 54 settings. Three earlier crossings occurred just above .80. At ICC .05 with 5% interval dropout, 70/30 allocation and normal sizes, 16 clusters replace the published 17, with mean power .80090. At ICC .05 with 15% dropout, balanced allocation and equal sizes, 15 replace 16, with mean power .80086. At ICC .15 with 15% dropout, 70/30 allocation and skewed sizes, 32 replace 33, with mean power .80190. At those three earlier counts, .360, .166 and .188 of masks fell below .80, respectively. Published threshold counts are compatible with a realized mask that crosses one cluster later. The largest extra count relative to the between–within reference was seven in R versus eight in Table 5. Neighboring counts and independent

covariance checks support the three earlier mean crossings (results/full/threshold-comparison.csv and results/full/threshold-scan.csv).

Disagreement record `threshold-mask`.

2.2 Oversampling for dropout

Adding clusters in proportion to cumulative participant dropout cleared 80% power in all six reported Table 3 scenarios. Dropout removes 15% or 45% of participants by the last occasion and 7.5% or 22.5% of observations across the four occasions under the design's schedule. Table 3 used 15, 22 and 29 clusters in the lower dropout arm and 19, 28 and 36 in the higher arm. The published power range was .83961–.90204 and the mean reconstructed range was .83938–.90192. At 19 clusters, ICC .05 and 45% cumulative dropout, the source's .88187 fell below the central 95% mask interval [.883690, .885691] from 10,000 masks, whose mean was .884912. In a separate saved diagnostic, dense and block generalized least squares (GLS) agreed at an adverse mask with power .881846. The source value is feasible in the lower tail, and the original mask is unknown. See results/full/oversampling-v2.csv, outputs/table3-gap-summary.csv, outputs/table3-independent-check.csv, and outputs/table3-allocation-summary.csv.

Disagreement record `table3-mask-tail`.

2.3 Planning with partially observed participants

The power calculations use all available occasions, including measurements contributed before a participant drops out. Kupzyk (2011, pp. 94–95, 99–100) also proposed in prose that planners add clusters in proportion to the fraction of *observations* lost across the study. We applied that rule to the six Table 3 settings using observation losses of 7.5% and 22.5%. We multiplied the baseline cluster counts by one plus the observation-loss fraction and rounded to the nearest cluster or upward. Across the 12 setting-and-rounding rows, mean powers ranged from 81.18% to 85.46% over 500 masks per row. All means exceeded 80% and were closer to it than those from the participant-loss rule in the same settings (results/full/observation-loss-oversampling.csv).

A completers-only analysis using listwise deletion would discard earlier observations from participants who later drop out. Its power is outside the design evaluated here. In the separate completers illustration, Kupzyk adds 15% to a target of 200 and obtains an enrollment of 230 (Kupzyk, 2011, p. 100). At 15% loss of initial enrollees, 230 gives $230(.85) = 195.5$ expected completers. The minimum integer enrollment to reach an expectation of 200 is $\lceil 200/.85 \rceil = 236$, with 200.6 expected completers (results/full/completers-arithmetic.json).

Disagreement record `completer-inflation`.

3 What we did ourselves and what we took from others

The reconstruction uses the population model, factor settings, original comparison targets and printed size frequencies from Kupzyk (2011, Chapters III–V and Appendices A–D). We extracted the published results, reconstructed the generator and tests, performed independent covariance and fitting checks, and compared fresh estimates with the published figures and tables. The model has occasions within people within randomized clusters. It has fixed effects for time, treatment and their interaction, independent Gaussian random intercepts and slopes at the person and cluster levels, and Gaussian occasion errors. The interaction is .5 under the alternative and zero in the null checks.

The 54 initial settings cross ICC .05, .10 and .15, 0%, 5% and 15% interval attrition, 50/50 or 70/30 allocation, and equal, normal-shaped or skewed cluster sizes. The ICC factor refers to the correlation of individual slopes within a cluster, and there are four occasions and 20 participants per cluster on average. The exemplary designs with unequal sizes use the source's printed integer frequency schedules (`outputs/normsizes.csv` and `outputs/skewsizes.csv`). The empirical simulations draw new cluster sizes from a normal distribution with mean 20 and standard deviation 2, round them to integers and constrain them to 15 through 25. For skewed sizes, they apply two transformations in alternation to draws above 20 and constrain the final sizes to 15 through 22 (`scripts/model.R`). Attrition removes a subset of the original participants at each interval without reentry. The empirical generator draws new outcomes and dropout masks. The exemplary calculation holds the variance components at their generating values. The initial cluster counts were 16, 22 and 28 as ICC increased, and the final exemplary grid used 13, 19 and 25. The six realized oversampling scenarios in Table 3 differ from the plans in Figure 5 (`source-values/figure-5.csv`).

Empirical power estimates use restricted maximum likelihood (REML) with a two-sided .05 interaction test. The source used SAS/IML and PROC MIXED. The SAS/STAT 15.2 manual documents the PROC MIXED REML default (SAS Institute, n.d., PROC MIXED statement, METHOD option). The R reconstruction profiles the REML objective and checks the fitted interaction and its standard error against lme4 (Bates et al., 2015) and nlme (Pinheiro et al., 2026), two R mixed-model implementations. A generator check compared 540 paired outputs and their random number states for the direct and reference implementations in R. The source's SAS calls `RANNOR(0)` and `RANUNI(0)` provide no fixed seeds for replay. See `rerun/validate_generator.R`, `rerun/validate_profile.R` and `results/engine-comparison.csv`.

Exemplary power conditions on the observed design matrix X and generating covariance V. With e selecting the treatment-by-time coefficient δ, the GLS variance is $v = e^{\mathsf{T}}(X^{\mathsf{T}}V^{-1}X)^{-1}e$. The noncentrality of the one-numerator-degree-of-freedom F test is δ^2/v. We checked the variance by three calculations using dense, block and reduced representations of the covariance (`results/full/fresh-gls-validation.csv`). The dropout calculations average power across independently sampled masks. The source's exemplary estimate is conditional on its realized design.

The formulaic calculation isolates the effect of the denominator degrees of freedom. In balanced complete designs with J clusters, n people per cluster and t occasions, Optimal Design uses $J-2$, while Kupzyk's between–within Equation 4.1 uses $Jnt-J-2$. At the initial baseline this gives 1,262 degrees of freedom and power .877206 under the between–within calculation, consistent with the source's approximately .88. The even balanced Optimal Design crossings are 16, 22 and 28, and allowing odd counts yields 16, 21 and 27. Formulaic between–within crossings are 13, 19 and 25. At the final baseline, formulaic and exemplary powers can differ even at a common denominator rule because the formulaic and realized design calculations differ. See `results/full/formulaic.csv` and Kupzyk (2011, pp. 83–89, 97, 103–104).

4 How well did the power estimates reproduce?

The 54 initial empirical estimates all agreed at the scale of the adjusted comparison. We ran 25,600 new draws per cell, for 1,382,400 successful fits. The source lost one fit in each of four cells. Reading the plotted values adds uncertainty to both the comparison and the estimates of the published Monte Carlo errors. We combined the two binomial Monte Carlo standard errors in quadrature and added the figure-reading allowance to the resulting band. A Monte Carlo standard error for a rejection fraction $\hat{p}$ from R independent draws is $\sqrt{\hat{p}(1-\hat{p})/R}$ (Morris et al., 2019). Two initial empirical cells exceeded the ordinary band of two combined standard errors plus reading error, and both were inside the two-sided 95% band

with a Bonferroni adjustment across 54 cells (Dunn, 1961). At ICC .10 with 70/30 allocation and normal sizes, the 5% attrition estimates were .792557 versus .780898, and the 15% estimates were .754458 versus .763672. The adjusted half-widths were about .0143 and .0141, respectively. These comparisons are checks at the resolution of the source and Monte Carlo estimates. See Kupzyk (2011, Figures 6–8) and `results/full/power-cell-comparison.csv`.

Exemplary power was also consistent at the resolution of the plotted values. We reconstructed all 54 initial and all 54 final settings. Three initial markers are hidden by other marks, leaving 51 readable initial comparisons. All 51 fit the adjusted bands, and all 54 final plotted comparisons fit their bands. The 18 no-dropout Table 1 values reproduce to five decimal places. With dropout, the variation among realized masks is distinct from the Monte Carlo error in estimating mean power over masks. A comparison of the source point with the mean uses the mask spread to account for the unknown source mask. See `source-values/figures-6-11.csv`, `source-values/table-1.csv`, `results/full/exemplary-v2.csv` and `results/full/power-cell-comparison.csv`.

Empirical power was usually below exemplary power. The source reports 41 of 54 initial cells below, and 39 of the 51 readable pairs are below. We find 40 of 54 below, with empirical minus exemplary power ranging from −.01661 to .00677. A separate 50,000-draw baseline run yielded .86842 against .877206 theoretical exemplary power, a deficit of .879 pp versus about .9 pp reported by Kupzyk (2011, pp. 106–107). See `results/full/empirical-exemplary-gaps.csv` and `results/full/empirical.csv`.

Kupzyk (2011, pp. 106–107) suggested that the gap would likely vanish with more simulation draws and that the two methods would then be equivalent. In a paired comparison on the same 50,000 balanced baseline outcomes, we held the reference critical value fixed and obtained power .87752 with the generating covariance and .86842 after refitting the covariance. The 43,421 refit rejections exactly reproduce the main follow-up count. Refitting lowered power by .00910 (paired Monte Carlo standard error .001068), supporting a covariance-refitting contribution to the baseline gap. Increasing the simulation count reduces Monte Carlo error around each rejection probability. This finite comparison is limited to the balanced baseline, and general convergence or equivalence remains unresolved (`results/full/baseline-paired.csv`).

Disagreement record `power-equivalence`.

The six Appendix C parameter tables agree at adjusted diagnostic precision. All 324 comparisons lie within their Bonferroni bands when the source's unavailable Monte Carlo standard errors are approximated by ours. The six parameters are the person and cluster intercept and slope variances, the slope ICC, and the time-by-treatment interaction. Boundary estimates, where a fitted variance equals zero, are distinct from failed fits. The initial baseline boundary fraction in R was .07266. The slope ICC is computed within each fit before averaging, so the average of the fitted ratios can differ from a ratio of averaged variances. See `results/full/appendix-c-comparison.csv` and `results/full/empirical.csv`.

5 Null calibration and denominator rules

The two Appendix D settings show that denominator rules can change both power and false positive rates. We used 25,600 draws under the null per setting. The corresponding alternative rates use the initial grid draws. We compared Optimal Design's $J - 2$ rule, between–within, containment, Satterthwaite and Kenward–Roger. The SAS manual documents the between–within and containment rules (SAS Institute, n.d., MODEL statement, DDFM options). We checked our Satterthwaite calculation (Satterthwaite, 1946) against the lmerTest implementation (Kuznetsova et al., 2017). Kenward–Roger also adjusts the covariance of the fixed effects. Its implementation was checked against pbkrtest (Halekoh and Højsgaard, 2014) (`results/full/kr-validation.csv`).

At baseline, between–within has a Type I error rate of .06500, compared with .06839 in Table D.1. Both exceed the nominal .05, as does containment. The reconstructed Kenward–Roger rate is .04625 with all variance components retained and .04891 when zero components are dropped. The source reports .05021. The reconstructed and source rates for all five rules fall within Bradley's liberal robustness interval [.025, .075] around .05 (Bradley, 1978), as applied by Kupzyk (2011, p. 105). The source's own two-standard-error interval for Satterthwaite and Kenward–Roger, [.04744, .05279], centers on .050115 rather than the reported .05021. The .000095 offset exceeds the .00001 rounding allowance. If the interval was symmetric at two standard errors, using the printed point in the variance formula implies about 26,658 draws, versus about 25,000 from most neighboring widths. These calculations leave the original interval procedure unresolved (`results/appendix-d-source-identity.csv`).

Disagreement record `appendix-d1-interval`.

The point classifications differ under the narrower robustness interval [.035, .065] that Kupzyk (2011, p. 105) attributes to Serlin (2000). The published between–within and containment rates of .06839 and .06811 exceed .065. The corresponding R estimates, .06500 and .0648828, lie at and just inside that endpoint. Both R rates are within .00012 of the cutoff, with Monte Carlo standard errors of about .00154, so the change in classification at the point estimates does not establish a difference in the underlying classification (`results/full/appendix-d-comparison.csv`).

Disagreement record `appendix-d1-serlin`.

At ICC .15 with skewed sizes, 15% interval attrition, and 70/30 allocation, the $J-2$ rule differs materially from Table D.2. Its power and Type I error at $J = 28$ and 26 degrees of freedom were .69594 and .04969, versus .67128 and .04279 in the source. The rerun's two-standard-error bands are approximately [.69019, .70169] and [.04697, .05240]. The published null interval [.04023, .04535] excludes the rerun point. A replay recovered 17,816 alternative and 1,272 null rejections at 26 degrees of freedom. Recomputing the tests on the same fits at 15 degrees of freedom gives .66953 and .04270. This sensitivity explains how a smaller effective denominator could produce both rates. The original denominator and fitted state are unresolved. See `results/full/appendix-d-comparison.csv`, `results/full/appendix-d2-df-sweep.csv` and `results/full/kenward-roger-summary.csv`.

Disagreement record `appendix-d2-od`.

6 Worked examples and simulation precision

The G*Power example reproduces to six decimals, and the two Optimal Design display values differ from the reconstructed formula. The published and recalculated G*Power values are both .584382. For the two Optimal Design examples they are .550898 versus .553630 and .422156 versus .425064. Computing the tail using either noncentral t or its square as a noncentral F gives the same recalculated values. The two differences, about .003 each, leave open the effect of a software convention or a different effective input. See Kupzyk (2011, Figures 1–3), `source-values/figures-2-3.csv` and `results/source-identities.json`.

Disagreement record `od-screen-example`.

The simulation-precision calculation supports the recommended draw counts. At power .80, a normal approximation with two standard errors on either side gives full width $4\sqrt{.8(.2)/R}$, so widths .01 and .005 require 25,600 and 102,400 draws. At 1,000 draws, the exact central binomial quantiles expressed as proportions are .775 and .824. Figure 4 shows two 500-draw illustrations. Our independent illustration uses a fixed standardized predictor and calibrates its population power to .80 at $n = 75$. Across 10,000 simulated power curves at each draw count, first 80% crossings ranged from 71 to 80 at 500 draws and 74 to 77 at 25,600 draws. These ranges illustrate Monte Carlo variability. The source toy model and its

two random streams are unavailable. See Kupzyk (2011, pp. 59–64), source-values/figure-4.csv and results/figure4-independent-summary.json.

7 Assumptions

7.1 Observed-count degrees of freedom

The between–within implementation extends the complete-design rule to $M - J - 2$, where M is the observed number of occasions summed over participants. This treats lost observations as reducing the within count. It equals Equation 4.1 for a complete balanced design. The extension is a reconstruction choice for the missing data settings (results/design-checks.csv).

7.2 Independent dropout realizations

The mask average estimates power under the reported sampling rule with independent realizations of missingness. The source's exemplary powers, oversampling cells, and cluster crossings are conditional on realized designs. The two estimands can differ even when they use identical covariance and test formulas. We retain the mask distribution to distinguish near-target threshold differences and lower-tail comparisons from fitting errors.

7.3 Unobserved source Monte Carlo variance

Published Appendix C summaries give means without the draw-level spread needed to compute their Monte Carlo errors. We use the corresponding rerun standard error as a proxy. The adjusted diagnostic bands depend on that approximation.

8 Ambiguities that were resolved

8.1 The Table 4 high-attrition column

Table 4 prints 10% per interval in the second attrition column. The inspected page confirms the label, and Figure 5 specifies 0%, 5% and 15%. Recomputing the 0-minus-15% contrasts from Table 1 gives 6.1692, 3.8722 and 3.1325 pp, which round to the column's 6.2, 3.9 and 3.1. We treat this as a source label error. It does not change the numerical findings (source-values/table-4.csv and results/source-identities.json).

Disagreement record `table4-high-attrition-label`.

8.2 The oversampling rule and the two reference counts

The printed oversampling counts follow the final baseline counts multiplied by one plus cumulative dropout and rounded to the nearest cluster. The proposed counts in Figure 5 follow a different grid starting at the initial Optimal Design counts. We compared power at the six realized counts in Table 3 and evaluated the four research questions at the final exemplary counts in Table 1 (results/source-identities.json).

8.3 Hidden markers and final figure values

Three exemplary markers in the initial figures are occluded. We left these observations missing in figure comparisons and used the independently reported full-grid count for the empirical-exemplary sign comparison. Table 1 provides the final numerical comparison and calibrates the reading of Figures 9–11 (`results/figure-table1-check.csv`).

9 What remains unresolved

The historical inputs needed for exact replay are the fitted SAS records, random states, and realized designs. The SAS syntax and generator checks fix the modeled distributions. Agreement across R fitting and GLS implementations constrains errors in our reconstruction. The published summaries leave the following comparisons unresolved at the scale indicated.

- For the three threshold differences, the neighboring-count scan and independent covariance check support the rerun crossings. Mask variation is the leading explanation. The realized masks at the published counts and their predecessors would allow a direct crossing comparison.
- For the 19-cluster Table 3 cell, an unusually adverse mask is feasible under the reconstructed rule. The allocation diagnostic and covariance cross-check do not identify the source configuration. The realized design and mask would distinguish sampling variation from allocation or rounding conventions.
- For Table D.2, a smaller effective denominator brings both rates near the printed pair and leads the explanations tested. Replay rules out a counting discrepancy between the R paths. The original per-fit test statistics, denominator degrees of freedom and rejection counts would distinguish a denominator convention from a different fitted distribution.
- For the Table D.1 intervals, transcription and ordinary rounding have been checked. The remaining candidates include a different rate, draw count, or interval procedure, with insufficient evidence to order them. A table-generation record with the original counts and interval formula would settle the calculation.
- For the two Optimal Design screens, equivalent tail formulas reproduce the same gaps. A software convention or a different effective input remains plausible. The original executable or saved project with effective inputs would distinguish these explanations. The Figure 4 seeds and generator would permit comparison with the exact plotted toy realizations.

10 Code and source availability

The University of Nebraska–Lincoln repository record identifies the dissertation and its canonical download. A complete archived copy of the university export was obtained on 2026-11-09. Its SHA-256 is `368dc575d9f744bf5ca5516873a633691d999fbdd5d19516994c64e10674896a`. It has 143 physical pages, with a ten-page offset from the printed Arabic numbering. The institutional record, direct PDF response and successful archival route are documented in `provenance/source-acquisition.json`. The copyright-restricted PDF is held separately. The source transcriptions and figure measurements are in the repository. See `source-inventory.json`.

SAS/IML, SAS PROC MIXED and Optimal Design 2.0 were unavailable for execution in this study. The engine record describes the R reconstruction and the loss of access to SAS-specific behavior and

Optimal Design output (notes/rerun-engine.json). The R and Python code and saved result tables are linked in Appendix A.

11 Next steps

Original design and fit records would turn several sensitivity comparisons into replay checks. A unit-level design and mask would let the same GLS calculation reproduce the published exemplary estimand at Table 3 and the three Table 5 thresholds. Per-fit test statistics and denominator degrees of freedom from Appendix D would let us recompute the extreme rates under the actual rule. An archived Optimal Design project would allow the reported and reconstructed effective inputs to be compared directly.

12 Numerical comparison at printed precision

The table distinguishes numerical agreement with a printed value from reconstruction scope. At the indicated precision, *match* means our value rounds to the published value, *close* means the gap is at most one unit in the last printed place, and *mismatch* means the gap exceeds that unit. All rows have limited scope because they use independently calculated powers or newly simulated data. The numerical labels use rounding, with Monte Carlo and mask comparisons reported separately above.

Table 2: Selected printed estimates. Sources are Kupzyk's Figures 2–3, Tables 1, 3 and 5 and Appendix D, and the corresponding reconstructed values.

Claim	Published	Ours	Precision	Numerical	Scope
G*Power example	.584382	.584382	6 decimals	match	limited
Optimal Design example, ICC 0	.550898	.553630	6 decimals	mismatch	limited
Optimal Design example, ICC .10	.422156	.425064	6 decimals	mismatch	limited
Final baseline exemplary power	.80081	.80081	5 decimals	match	limited
Table 3, 19 clusters	.88187	.88491	5 decimals	mismatch	limited
Threshold, cell 11	17	16	1 cluster	close	limited
Threshold, cell 13	16	15	1 cluster	close	limited
Threshold, cell 54	33	32	1 cluster	close	limited
D.1 between–within Type I error	.06839	.06500	5 decimals	mismatch	limited
D.1 Kenward–Roger Type I error	.05021	.04625	5 decimals	mismatch	limited
D.2 Optimal Design power	.67128	.69594	5 decimals	mismatch	limited
D.2 Optimal Design Type I error	.04279	.04969	5 decimals	mismatch	limited

The worked example matches for the noncentral t calculation and differs for the two software displays.

At the final baseline the fixed design yields the same printed power. At 19 clusters the source value is in the far lower tail of the reconstructed mask distribution. All three listed thresholds differ by exactly one cluster. The two D.1 rate gaps exceed the rounding unit. All baseline rates lie inside the broad Bradley practical band [.025, .075]. The between–within and containment point estimates change classification under the narrower Serlin band [.035, .065], with the R values on or just inside .065. Both D.2 gaps exceed the rounding unit and the rerun two-standard-error bands.

A Artifacts and regeneration

Use the repository's `master` branch. Set the study path from the repository root before using either regeneration route.

```
study=formalization/kupzyk-power-assumptions
```

R, numerical reconstruction. Run

```
sh "$study/rerun/reproduce.sh" /tmp/kupzyk-reproduction 4
```

R 4.5.3 supplied random draws, distribution functions and numerical optimization (R Core Team, 2026). Rcpp 1.1.2 compiled the C++ profile evaluator (Eddelbuettel and François, 2011). The fitting and denominator checks used lme4 2.0-6 (Bates et al., 2015), nlme 3.1-170 (Pinheiro et al., 2026), lmerTest 3.2-1 (Kuznetsova et al., 2017), pbkrtest 0.5.5 (Halekoh and Højsgaard, 2014), and Matrix 1.7-5 for rank calculations (Bates et al., 2026). Python 3.14.10 aggregated and compared results (Python Software Foundation, n.d.). The independent precision calculation used NumPy 2.5.4 for arrays (Harris et al., 2020) and SciPy 1.18.1 for numerical roots and distributions (Virtanen et al., 2020). PyYAML 6.0.3 parsed the evidence records (PyYAML contributors, n.d.). PyMuPDF 2.1 extracted source-figure geometry, and PyMuPDF 1.26.6 extracted annotations from the replication PDF for link review (PyMuPDF contributors, n.d.). The environment is recorded in `provenance/rerun-runtime.txt`. See the prerequisites and checkpoints in `rerun/README.md` and the exact command chain in `rerun/reproduce.sh`. The full empirical grid schedules 128 batches of 200 draws per setting, plus the 50,000-draw follow-up and two null cases. The following paths are relative to the selected run directory. Each link opens the corresponding checked-in result.

- `rerun/formulaic.R` writes `formulaic.csv`.
- `rerun/precision_demo.py` writes `figure4-independent-summary.json`.
- `rerun/validate_profile.R` writes `engine-comparison.csv`.
- `rerun/check_oversampling.py` writes `comparisons/observation-loss-oversampling.csv` and `comparisons/completers-arithmetic.json`.
- `rerun/baseline_paired.R` writes `baseline-paired.csv`.

Reuse the checkpoint directory for a resumed identical run. Use a new directory after changes to scientific inputs, versions or seeds.

S, source measurements. Retrieve the archived PDF recorded in `provenance/source-acquisition.json` and verify its SHA-256. Poppler 26.10.0 extracts layout text from the source (Poppler Project, 2026). Use the source audit instructions in `rerun/README.md`. This step regenerates the source transcriptions and calculations from the archival PDF. Route R uses the checked-in transcriptions and can run without the archival file.

The report rebuild uses the saved results. With Tectonic 0.17.0 (Tectonic contributors, n.d.) and its LaTeX bundle, run the following command from the repository root.

```
(cd "$study/paper" && tectonic kupzyk-replication.tex)
```

The bibliography is embedded in the editable source. This rebuild writes the PDF alongside the source.

Table 3: Evidence by claim and regeneration route. Published page locators refer to the printed dissertation.

Claim	Evidence	Route
Chapter I motivation (pp. 1–6)	notes/claim-inventory.md	S
Chapter II software examples	results/source-identities.json	S
Figure 4 simulation precision	results/figure4-independent-summary.json	R
Chapter III crossed design	source-values/figure-5.csv	S
Initial empirical grid	results/full/empirical.csv	R
Initial exemplary grid	results/full/exemplary-v2.csv	R
Final exemplary grid	results/full/power-cell-comparison.csv	R
Optimal Design formula	results/full/formulaic.csv	R
Between–within formula	results/full/fresh-gls-validation.csv	R
Chapter IV results	results/full/comparison-summary.json	R
Attrition contrasts	results/full/paired-contrasts.csv	R
Table 3 and observation-loss oversampling	results/full/oversampling-v2.csv, results/full/observation-loss-oversampling.csv	R
Allocation contrast	results/full/paired-contrasts.csv	R
Size contrasts	results/full/paired-contrasts.csv	R
ICC moderation	results/full/comparison-summary.json	R
Predicted loss ordering	source-values/table-1.csv, results/full/paired-contrasts.csv	S, R
Table 5 thresholds	results/full/threshold-comparison.csv	R
Chapter V interpretation	results/full/appendix-d-comparison.csv, results/full/completers-arithmetic.json	R
50,000-draw baseline	results/full/empirical.csv, results/full/baseline-paired.csv	R
Empirical–exemplary gap	results/full/empirical-exemplary-gaps.csv	R
Appendix A generator and REML	rerun/validate_generator.R, results/engine-comparison.csv	R
Appendix B covariance and sizes	results/full/fresh-gls-validation.csv, outputs/normsizes.csv, outputs/skewsizes.csv	R, S
Appendix C fitted means	results/full/appendix-c-comparison.csv	R
Appendix D.1 baseline	results/full/kenward-roger-summary.csv	R
Appendix D.2 extreme setting	results/full/appendix-d2-df-sweep.csv	R
Practical planning ordering	results/full/paired-contrasts.csv, results/full/threshold-comparison.csv	R

B Compliance ledger

Requirement	Status	Evidence
Source verification	followed	`provenance/source-acquisition.json` records the complete archival copy.
Design coverage	followed	`REPLICATION-PLAN.yaml` and `results/full/comparison-summary.json` account for the crossed grid and follow-ups.
Original engine	departed	SAS and Optimal Design were unavailable. The independent R checks and limits are described in `notes/rerun-engine.json`.
Exact source streams	departed	The published code supplies zero-valued seed arguments and the unit-level masks are unavailable. New streams cover all settings.
Comparisons and uncertainty	followed	`results/full/power-cell-comparison.csv` separates reading allowances, Monte Carlo error and mask spread.
Citation scope	followed	The `citation-inventory.json` maps methods and software to their citations and accounts for all earlier references.
Discrepancy record	followed	`DISAGREEMENTS.yaml` records the checks and unresolved cases.

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